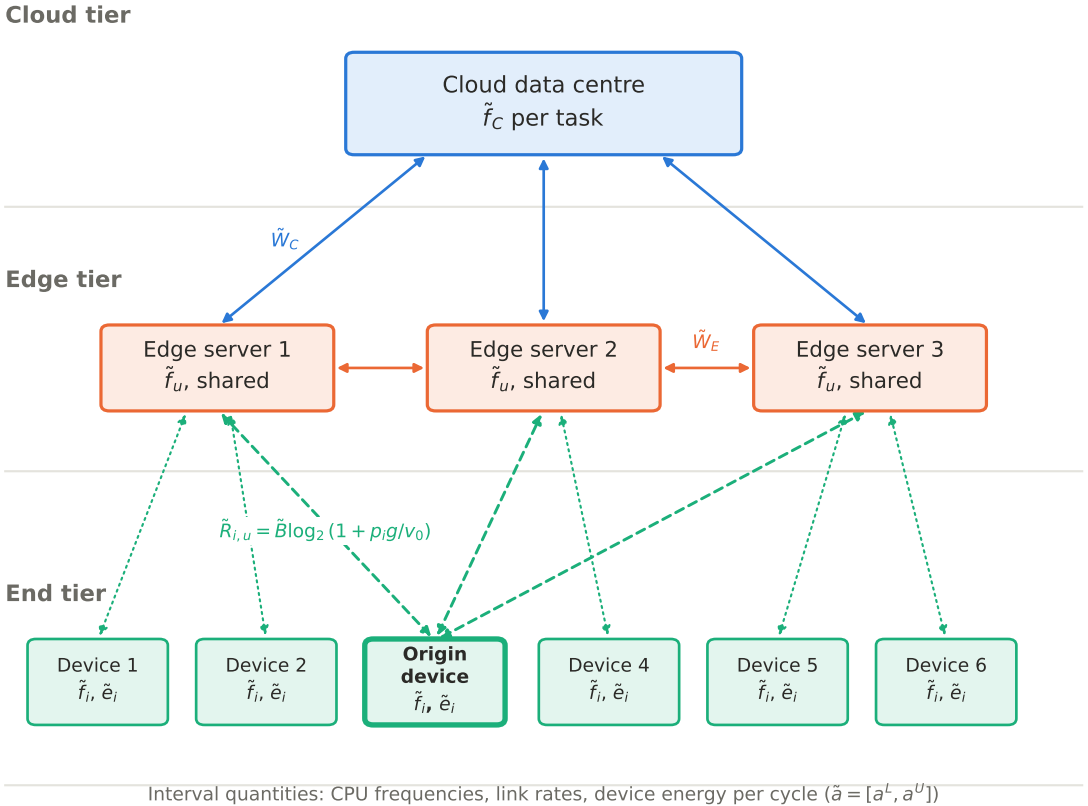
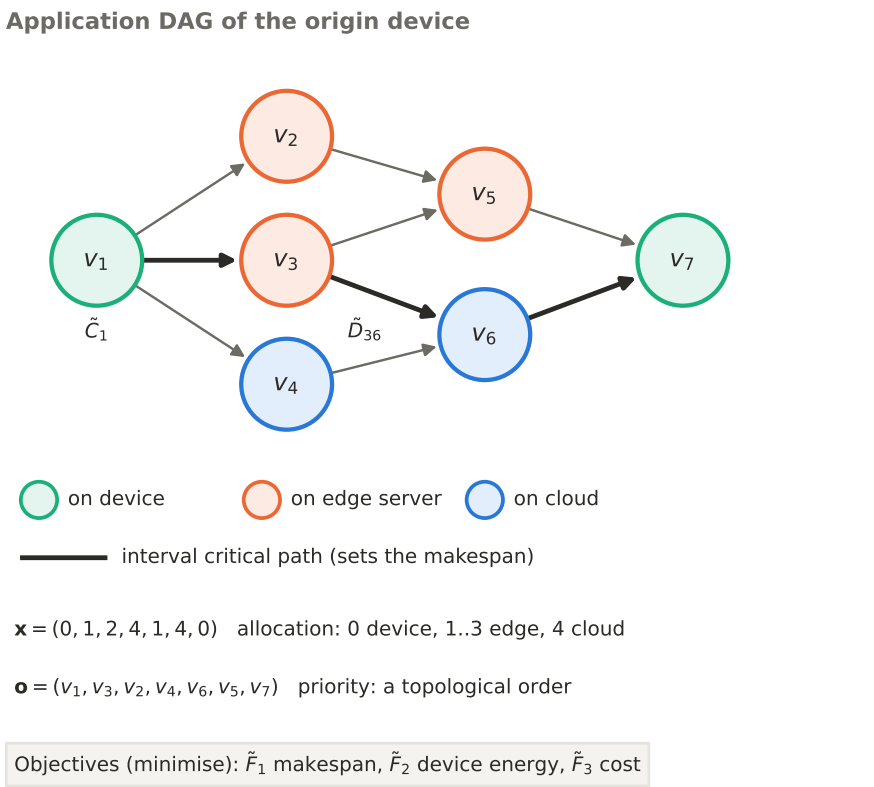

Cloud tier
Cloud data centre
$\tilde{f}_C$ per task
$\tilde{W}_C$
Edge tier
Edge server 1
$\tilde{f}_u$, shared
Edge server 2
$\tilde{f}_u$, shared
$\tilde{W}_E$
Edge server 3
$\tilde{f}_u$, shared
$\tilde{R}_{i,u} = \tilde{B}\log_2(1 + p_i g / v_0)$
End tier
Device 1
$\tilde{f}_i$, $\tilde{e}_i$
Device 2
$\tilde{f}_i$, $\tilde{e}_i$
Origin device
$\tilde{f}_i$, $\tilde{e}_i$
Device 4
$\tilde{f}_i$, $\tilde{e}_i$
Device 5
$\tilde{f}_i$, $\tilde{e}_i$
Device 6
$\tilde{f}_i$, $\tilde{e}_i$
Interval quantities: CPU frequencies, link rates, device energy per cycle ($\tilde{a} = [a^L, a^U]$)


Application DAG of the origin device
v_1
v_2
v_3
v_4
v_5
v_6
v_7
$\tilde{C}_1$
$\tilde{D}_{36}$
on device
on edge server
on cloud
interval critical path (sets the makespan)
$\mathbf{x} = (0, 1, 2, 4, 1, 4, 0)$ allocation: 0 device, 1..3 edge, 4 cloud
$\mathbf{o} = (v_1, v_3, v_2, v_4, v_6, v_5, v_7)$ priority: a topological order
Objectives (minimise): $\tilde{F}_1$ makespan, $\tilde{F}_2$ device energy, $\tilde{F}_3$ cost